

TellCo Acquisition Analysis

User overview, engagement, experience and satisfaction analysis of one week of xDR data, with an investment view

Prepared by **Srishti Sharma** · NextHikes IT Solutions

Dataset: telcom_data (2).xlsx · 150,001 xDR session rows × 55 columns · 106,856 customers · sessions ending 24–30 April 2019

106,856 Customers analysed	148,935 Sessions after cleaning	1 week Not a month (24–30 Apr 2019)	0.71% Rows removed (no customer ID)	6.0% Sessions with no known handset
22.8% High-engagement customers (≈37% of traffic)	34.6% Good-throughput customers	40.4% Low-throughput customers	63.2% Customers with TCP data estimated in every session	0.992 Model R ² (recovers a formula, not a forecast)

1. Executive summary

An investor is considering buying TellCo, a mobile service provider in the Republic of Pefkakia. TellCo's owners have shared financial information but have never analysed their own system data. This report analyses TellCo's xDR (data-session) records to show who the customers are, how much they use the network, how good their connection is, and what that means for a purchase decision.

Two points shape everything that follows. **First, the file covers one week, not a month:** every session ends between 24 and 30 April 2019, so all results describe that week. **Second, the file holds no revenue, cost, churn or survey data,** so it can describe behaviour but cannot say whether TellCo is profitable or undervalued.

Recommendation: continue due diligence — do not buy or walk away on this data alone

Growth potential: cautiously positive, but unproven. TellCo has a large, identifiable customer base, a sizeable high-usage group and network gaps that look fixable, not a business in structural decline. But one week of usage cannot show growth, profitability or customer loss.

Before a buy / do-not-buy decision: obtain financial data (revenue, ARPU, cost, churn), a longer and evenly sampled data window, outcome data (surveys, complaints, churn), and cell-level network data. Section 9 lists the full set.

Key findings

- **Customers.** 106,856 customers generated 148,935 sessions, 1.39 sessions each; 72.7% had a single session. The average customer moved 691 MB in the week (median 618 MB).
- **Devices.** Apple (39.9% of sessions), Samsung (27.2%) and Huawei (23.1%) account for 90.2%. The single most common “handset”, the Huawei B528S-23A (13.2% of sessions), is a home internet router, not a phone. 6.0% of sessions have no usable device information.

- **Applications.** Gaming and Other make up 93.8% of the seven application counters. That is a share of the counters, not of TellCo's total traffic, because the file's totals are built inconsistently (Section 3). Social Media and Email are under 0.3% each.
- **Engagement.** 22.8% of customers are highly engaged and carry about 37% of traffic. A second group (23.1%) uses no more data than the low group (54.1%) but keeps connections open about 2.5 times longer. Traffic is moderately concentrated: the top 10% of customers carry about 26% of volume.
- **Experience.** Throughput is the one network measure we can trust. 34.6% of customers have good throughput (home-router users are over-represented), 40.4% have low throughput, and 25.1% sit in a group whose latency is mostly unmeasured. TCP retransmission data is largely estimated and should not be used for investor conclusions.
- **Satisfaction.** The file has no real satisfaction data. The brief's score is a behavioural proxy, and the model that predicts it ($R^2 = 0.992$) recovers a formula rather than forecasting customer behaviour.
- **Method matters.** The brief's rule of replacing outliers with the average changes 15.0% of customers, who carry 29.9% of raw traffic. It hides the heaviest users, so we report untreated figures alongside for any question about heavy users or concentration.

2. Business need and approach

The investor buys undervalued assets and uses data to find where a business can be refocused to raise profitability. The brief asks for four analyses, a recommendation on TellCo's growth potential and a view on whether to buy. This report covers the analysis and the recommendation; the dashboard, slide deck and engineering deliverables accompany it (Section 12).

TASK	QUESTION ANSWERED	WHAT WE DID
1. USER OVERVIEW	Who are the customers and what do they use?	Ranked handsets and manufacturers, summarised each customer's sessions, duration and application volumes, and tested how the pieces relate.
2. USER ENGAGEMENT	Who uses the network most?	Ranked top customers, then grouped customers into three engagement segments (k-means, $k = 3$).
3. USER EXPERIENCE	How good is the connection?	Summarised retransmission, latency and throughput per customer and grouped customers into three experience segments.
4. USER SATISFACTION	Who is likely to be satisfied?	Built engagement and experience scores, averaged them into a satisfaction proxy, modelled it and grouped customers into two segments.

How the numbers were checked. All figures in this report come from a fresh end-to-end run of the final notebook on the supplied file, and were compared with the notebook's own results. Where an earlier draft of this report differed (for example, calling the data "one month"), this version follows the verified results.

3. Data quality: what we found and fixed

The data needed substantial checking before analysis. Rather than silently deleting or smoothing problems away, each one was measured, treated in a stated way and carried through the later stages.

DATASET AT A GLANCE	VALUE
RAW ROWS × COLUMNS	150,001 × 55 (54 after dropping a duplicate duration column)
ROWS REMOVED (NO CUSTOMER ID)	1,066 (0.71% of rows and 0.71% of traffic), including one row of column averages
SESSIONS ANALYSED	148,935
UNIQUE CUSTOMERS (MSISDN)	106,856; 1.39 sessions each; 72.7% have one session
PERIOD COVERED	Sessions end 24–30 April 2019 (14.9–17.4% per full day)
FULLY DUPLICATED ROWS	0

ISSUE	WHAT WE FOUND	WHAT WE DID	WHY IT MATTERS
PERIOD IS ONE WEEK	The brief says a month, but every session ends 24–30 April 2019.	Relabelled every result as one week.	No trend, seasonality or monthly claim is possible.
MISLABELLED DURATION	“Dur. (ms)” is really seconds; a second column holds the true milliseconds.	Kept the seconds column, renamed it, dropped the duplicate.	Avoids a 1,000-fold error in every duration figure.
24-HOUR PILE-UP	17.2% of sessions last exactly 24 hours and 44.4% last longer.	Treated duration as “connection open time”, not usage time.	Duration partly reflects system behaviour, not customers.
SESSION ID IS NOT A KEY	Bearer Id is blank in 991 rows and repeats across different customers.	Counted sessions per customer ID (MSISDN) only.	Avoids wrongly merging or splitting customers.
UNKNOWN HANDSETS	6.0% of sessions say “undefined” or are blank.	Labelled “Unknown” rather than guessing the most common handset.	Prevents inflating the top handset’s share.
TOTALS DO NOT ADD UP	Total download excludes the “Other” download counter, but total upload includes it, so the seven app counters sum to 185% of the total.	Reported app shares as shares of the counters, not of total traffic.	App percentages must not be quoted as shares of TelCo’s traffic.
MISSING NETWORK DATA	TCP retransmission is missing for 59–65% of sessions, HTTP volume for about 54%, latency (RTT) for about 18.6%.	Filled with the column average as the brief requires; kept a flag on every filled value.	The filled TCP value is about 36 times the typical measured value, so filled sessions look worse than real ones.
OUTLIERS	The brief’s average-replacement rule changes 15.0% of customers, who carry 29.9% of raw traffic.	Used the treated table as primary; reported untreated and capped versions alongside.	Heavy users are hidden, and concentration looks lower than it is.
SUSPICIOUS COUNTERS	YouTube and Netflix have almost identical averages, as do Gaming and Other.	Flagged; no app-level conclusion is used commercially.	Counters may be evenly split or synthetic; confirm with TelCo.
POSSIBLE COUNTER CEILING	The largest TCP value sits within 0.013% of the 32-bit limit (4,294,967,295).	Flagged as a possible cap; not treated as proof.	The extreme end of TCP data may be understated.

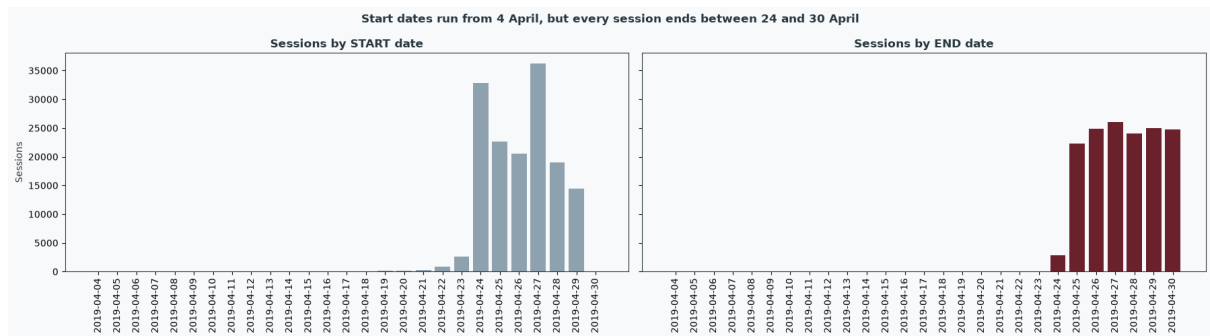


Figure 1. Sessions by start date (left) and end date (right). Every session ends 24–30 April 2019.

4. Task 1: User overview

4.1 Handsets and manufacturers

Handsets are counted by session, as the brief asks. “Unknown” is a placeholder for sessions with no device information and appears fourth; it is not a device.

RANK	HANDSET	SESSIONS	SHARE OF SESSIONS
1	Huawei B528S-23A	19,727	13.2%
2	Apple iPhone 6S (A1688)	9,413	6.3%
3	Apple iPhone 6 (A1586)	9,012	6.1%
4	Unknown (no device recorded)	8,931	6.0%
5	Apple iPhone 7 (A1778)	6,304	4.2%
6	Apple iPhone Se (A1723)	5,176	3.5%
7	Apple iPhone 8 (A1905)	4,985	3.3%
8	Apple iPhone Xr (A2105)	4,562	3.1%
9	Samsung Galaxy S8 (Sm-G950F)	4,480	3.0%
10	Apple iPhone X (A1901)	3,810	2.6%

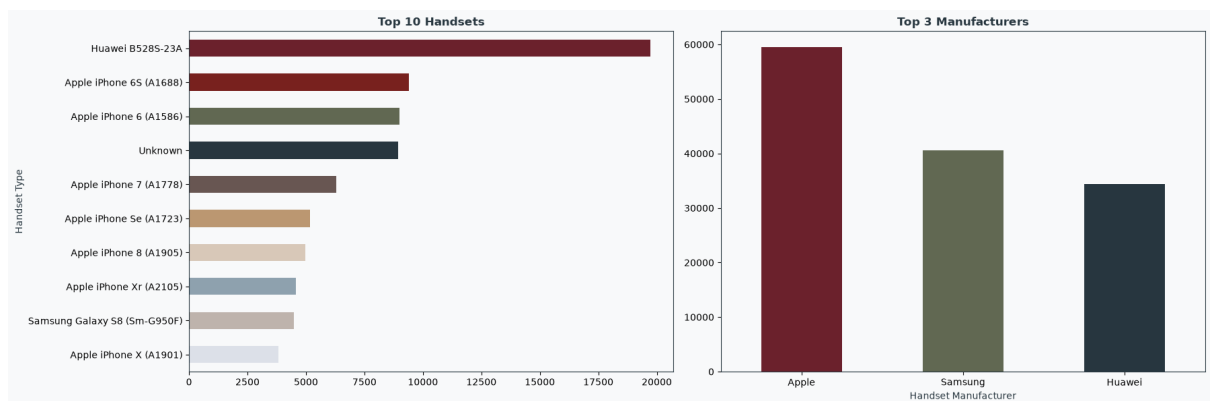


Figure 2. Top 10 handsets (left) and top 3 manufacturers (right), by sessions.

Apple leads with 59,464 sessions (39.9%), then Samsung with 40,579 (27.2%) and Huawei with 34,366 (23.1%), together 90.2%. Counted by customer instead of session, the ranking holds: Apple 39.9%, Samsung 28.8%, Huawei 20.3%.

MANUFACTURER	TOP 5 HANDSETS	SESSIONS
APPLE	Apple iPhone 6S (A1688)	9,413
	Apple iPhone 6 (A1586)	9,012
	Apple iPhone 7 (A1778)	6,304
	Apple iPhone Se (A1723)	5,176
	Apple iPhone 8 (A1905)	4,985
SAMSUNG	Samsung Galaxy S8 (Sm-G950F)	4,480
	Samsung Galaxy A5 Sm-A520F	3,708
	Samsung Galaxy J5 (Sm-J530)	3,682
	Samsung Galaxy J3 (Sm-J330)	3,464
	Samsung Galaxy S7 (Sm-G930X)	3,176
HUAWEI	Huawei B528S-23A	19,727
	Huawei E5180	2,074
	Huawei P20 Lite Huawei Nova 3E	2,018
	Huawei P20	1,479
	Huawei Y6 2018	997

The Huawei B528S-23A is the most frequent handset for 10,615 customers (9.9% of all customers), who average 1.86 sessions against 1.39 overall. It looks like a separate home-broadband segment inside Huawei's count. That is an inference from the model name, and it should be confirmed with TelCo.

Recommendation to marketing (a hypothesis to test, since there is no revenue data)

- 1. Test upgrade or trade-in offers** on the large iPhone 6S, 6, 7, SE and 8 base. These are previous-generation phones for April 2019, not necessarily old.
- 2. Market the B528S-23A group separately** as home-broadband customers, with their own plans, and analyse their network use apart from phones.
- 3. Treat Samsung demand as spread** across mid-range models (S8, A5, J5, J3, S7), where price-led campaigns are the natural test.
- 4. Ask TelCo why 6.0% of sessions have no device recorded.**

4.2 Application usage

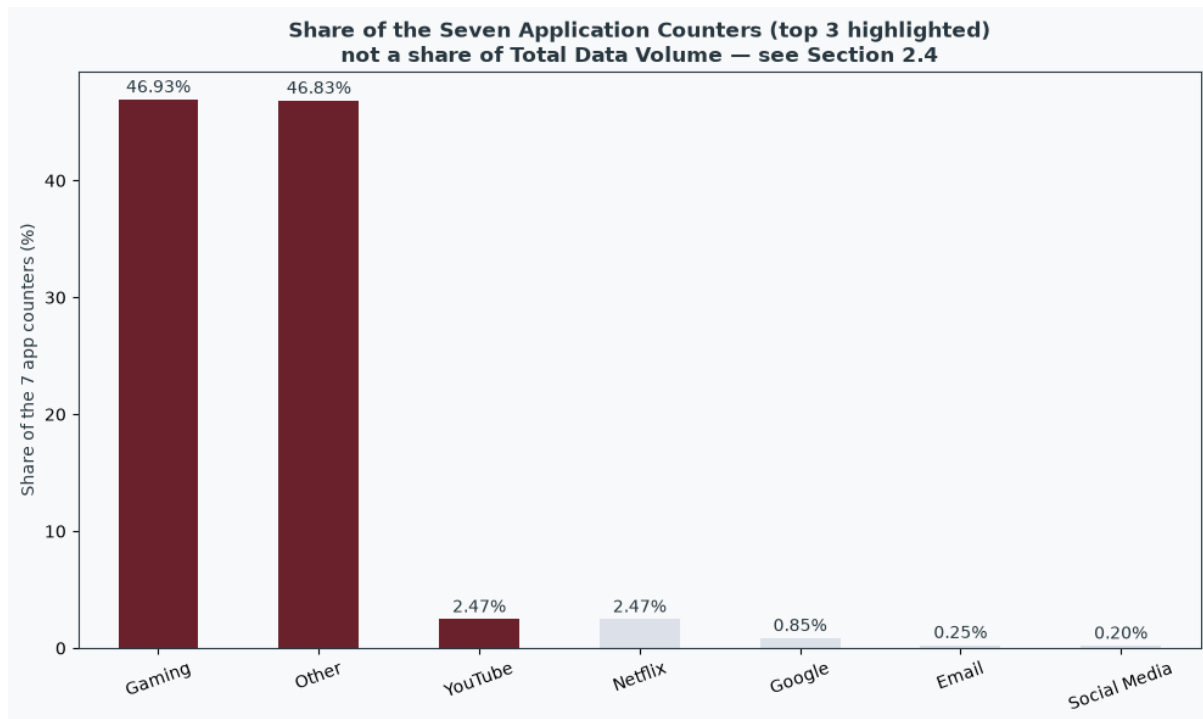


Figure 3. Share of the seven application counters (not of total traffic).

APPLICATION	SHARE OF THE SEVEN COUNTERS
GAMING	46.93%
OTHER (CATCH-ALL BUCKET)	46.83%
YOUTUBE	2.47%
NETFLIX	2.47%
GOOGLE	0.85%
EMAIL	0.25%
SOCIAL MEDIA	0.20%

Among named applications the top three are Gaming, YouTube and Netflix, with YouTube and Netflix effectively tied. Gaming averages about 87% of each customer’s recorded total volume. Its near-perfect correlation with total volume (0.98) is therefore mostly a size effect: it is most of the total by construction.

So what: Do not tell the investor that “gaming drives heavy users”. Gaming’s weight may reflect real preference or how the file defines volume, so ask TelCo how “Gaming” and “Other” are defined before using app shares in pricing or marketing.

4.3 Typical customer and spread

PER CUSTOMER (UNTREATED)	MEAN	MEDIAN	STD DEV	MAX
SESSIONS	1.39	1	0.81	18
DURATION (SECONDS OPEN)	146,167	102,740	186,359	18,553,754
TOTAL DATA (MB, DL + UL)	691	618	491	8,846

Every measure is right-skewed: the average sits above the median because a minority of heavy users pull it up, so the median describes the typical customer better. Total duration is the extreme case. The longest customer's 18.6 million seconds is about 215 days, longer than the observation window, because it adds up overlapping open connections across 17 sessions. Read duration as "connection-seconds", not time on device.

So what: Network load and revenue are driven by the heavier tail, not the typical customer. The median customer makes one session and moves about 618 MB in the week.

4.4 How applications relate to total traffic

RELATIONSHIP WITH TOTAL DATA VOLUME	TREATED (BRIEF)	UNTREATED
GAMING	0.98	1.00
OTHER SIX APPLICATIONS	0.33–0.40	0.62–0.71
OTHER SIX, PER SESSION (REMOVES THE EFFECT OF CONNECTING MORE OFTEN)	0.06–0.09	—
PAIRS OF APPLICATIONS WITH EACH OTHER	0.31–0.42	0.59–0.74

Once we remove the effect of simply connecting more often, the other six applications show almost no link to volume. Customers with more sessions use more of everything, so the moderate correlations mostly reflect session frequency, not application preference. YouTube and Netflix are the most closely linked pair and Gaming and Other the least.

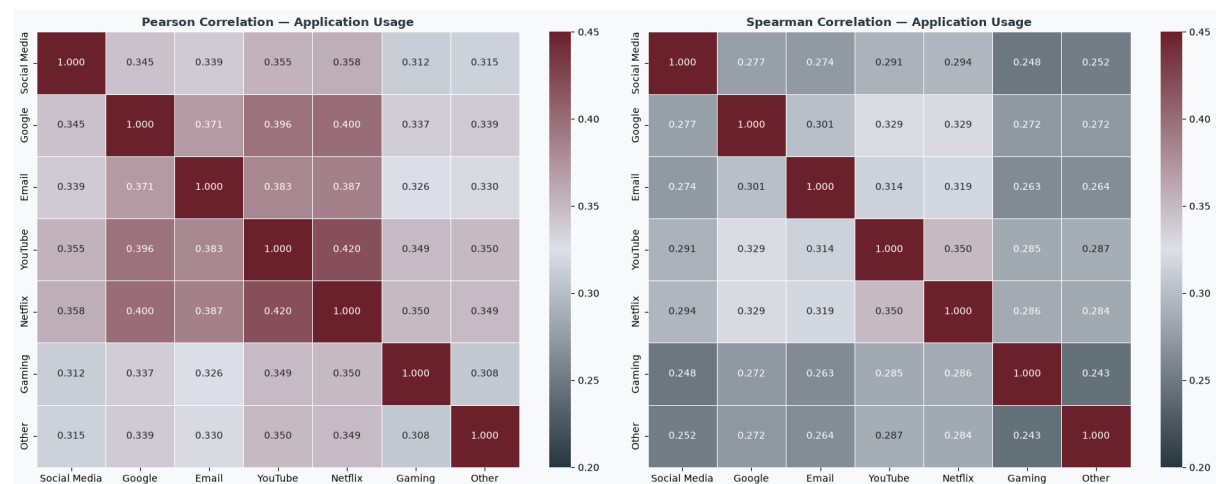


Figure 4. Application correlations, treated table: Pearson (left) and Spearman (right).

What the principal component analysis shows

- **One “general usage” factor.** It explains 44.6% of the variation, and every application loads positively: customers range from generally light to generally heavy.
- **Weak compression.** PC1 explains 44.6% of the variation; five components are needed for 80% and six for 90%. On untreated data PC1 explains 70.7% and three components reach 80%, so much of the weak compression comes from the mean-replacement treatment.
- **Do not over-read the second and third factors.** The second separates Gaming from Other, consistent with their weak link, but the loadings are unstable.
- **The data suits this analysis well** (sampling adequacy 0.88), and it is a description of one week only.

So what: Customers are best segmented by how much they use, not by which applications they prefer. Manufacturers differ in level, not in mix: Samsung customers use 5.6–6.9% less than Apple customers on every application, with an identical application ranking.

4.5 Duration deciles

Customers were ranked by total session duration and split into ten equal groups. In the treated table, total data rises from 508 MB per customer in the shortest-duration decile to 791 MB in the longest, but not steadily. The top five deciles carry 56.0% of traffic (60.4% untreated), and the longest decile carries 12.8% (20.1% untreated).

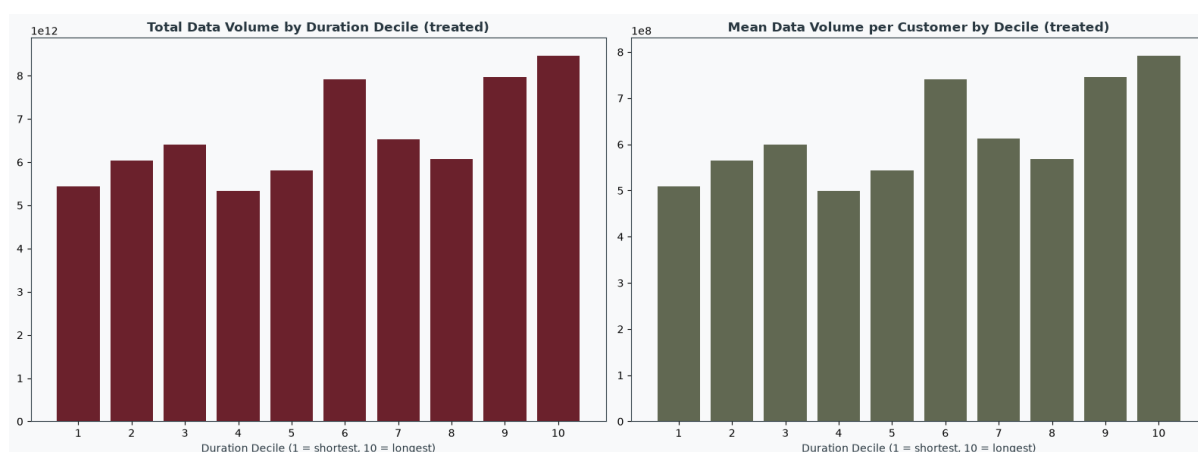


Figure 5. Total data (left) and mean data per customer (right) by duration decile, treated table.

Two quirks explain the dips. About 12,177 customers (11.4%) have a duration between 85,800 and 86,400 seconds, right at the 24-hour mark, which depresses the fourth decile. In the treated table, 8,052 customers share the single replacement duration of 114,380 seconds, which scrambles the boundary between the sixth and seventh deciles.

So what: The longest-duration customers carry more traffic than the treated table suggests. Use the untreated view for any heavy-user question.

4.6 What the brief's mean replacement changes

MEASURE	TREATED (BRIEF)	UNTREATED
CUSTOMERS CHANGED / SHARE OF RAW TRAFFIC THEY CARRY	16,033 (15.0%) / 29.9%	—
GINI COEFFICIENT (0 = EQUAL, 1 = ONE CUSTOMER)	0.304	0.356
TRAFFIC CARRIED BY THE TOP 10% OF CUSTOMERS	21.2%	25.9%
TRAFFIC CARRIED BY THE TOP 1% OF CUSTOMERS	2.6%	4.5%
STRENGTH OF APPLICATION PAIRS	0.31–0.42	0.59–0.74
VARIATION EXPLAINED BY THE FIRST PCA FACTOR	44.6%	70.7%

Concentration is moderate, not extreme, in every version. TellCo does not appear to depend on a razor-thin slice of customers, and the untreated numbers are the better estimate of the real picture.

5 Task 2: User engagement

Engagement is measured three ways: how often a customer connects (sessions), how long connections stay open (duration) and how much data they move (traffic). Top-10 lists use untreated data, because mean replacement overwrites exactly the extreme customers these lists exist to find.

5.1 Top 10 customers per engagement metric

#	BY SESSIONS: CUSTOMER	SESSIONS	BY DURATION: CUSTOMER	SECONDS OPEN	BY TRAFFIC: CUSTOMER	GB
1	33626320676	18	33625779332	18,553,754	33614892860	8.85
2	33614892860	17	33614892860	9,966,898	33760536639	8.51
3	33625779332	17	33760536639	9,279,434	33625779332	8.50
4	33659725664	16	33626320676	8,791,927	33626320676	7.97
5	33675877202	15	33667163239	8,744,914	33675877202	7.89
6	33760536639	15	33662840755	6,614,270	33659725664	7.71
7	33667163239	13	33664693736	6,288,730	33666464084	7.31
8	33603127838	12	33603127838	6,287,761	33760413819	7.13
9	33604515716	12	33667456716	5,649,882	33664712899	6.87
10	33627080969	12	33760413819	5,321,667	33698792269	6.54

The busiest customer connects 18 times against a population median of 1. Several customers appear on all three lists, but the top-traffic and top-duration customers are different people, so engagement is not one number. Ties matter at the cut-off: five customers have exactly 12 sessions, so the sessions list is one valid top 10, not the only one.

5.2 Three engagement segments (k-means, k = 3)

Each metric was standardised so no single scale dominates, and customers were grouped with k-means. The grouping is very stable across random starts (agreement 0.998). Segments are numbered from least to most engaged.

SEGMENT	CUSTOMERS	AVG SESSIONS	AVG DURATION (HOURS OPEN)	AVG DATA (MB)	SHARE OF TRAFFIC
LOW ENGAGEMENT	57,834 (54.1%)	1.02	19.3	506	44.3%
LONG-HOLD, LIGHT TRAFFIC	24,654 (23.1%)	1.03	48.0	491	18.3%
HIGH ENGAGEMENT	24,368 (22.8%)	2.17	45.0	1,011	37.3%

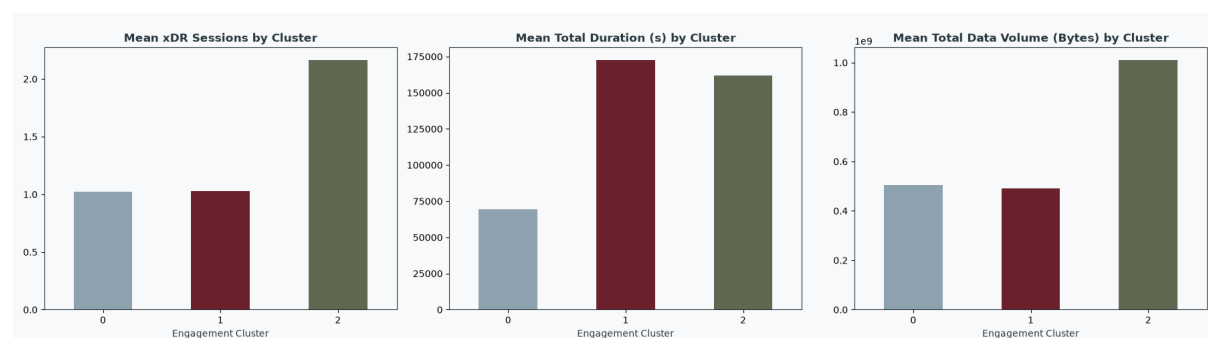


Figure 6. Average sessions, duration and data by engagement segment (treated table).

- **High engagement (22.8%)** leads on sessions (2.17) and traffic (about 1.0 GB) and carries about 37% of treated traffic.
- **Low engagement (54.1%)** has about one session, 19 hours of open connection and 506 MB of data. It is the reference point for the engagement score.
- **Long-hold, light traffic (23.1%)** moves the same data as the low group (491 MB) but keeps connections open about 2.5 times longer (48 hours). This is connections held open, not heavier use, and the 24-hour caveat applies.

So what: The segments are not a simple low–medium–high ladder. The first two differ in how long connections stay open, not in how much customers use the network. The valuable group to protect is the 22.8% High-engagement segment.

Is three the right number of segments?

Not on the data alone. $k = 2$ scores better on every internal quality measure (separation score 0.51 against 0.34 for $k = 3$); the “elbow” method points to $k = 5$. The third segment mostly splits the low group by connection length. We use $k = 3$ as the brief requires, but test $k = 2$ against real business outcomes before committing to three segments.

The segments depend on the outlier treatment

Under the brief’s mean replacement, the ten customers with the most sessions (12–18) are recorded as one-session customers and fall in the low-engagement segment. When extremes are capped rather than replaced, all ten fall in the top segment, and the segments form a clean ladder (sessions 1.0 / 1.9 / 3.6). Agreement between the two versions is only 0.44. The treated result stays primary, but conclusions about the heaviest users and the long-hold group depend on the treatment.

The full minimum, maximum, average and total of each metric per segment are in Appendix B, and the top 10 customers for each application are in Appendix A.

6. Task 3: User experience

Experience is measured by three network measures per customer: TCP retransmission (data re-sent after loss), round-trip time (RTT, the delay, in milliseconds) and throughput (speed, in kbps). Higher throughput is better; lower RTT and retransmission are better. Handset type is each customer’s most frequent handset.

How reliable are these measures?

Throughput is reliable: it has almost no missing data.

RTT is partly estimated: about 17% of customers have RTT estimated in every session.

TCP retransmission is mostly estimated: 63.2% of customers have an estimated value in every session, 52.4% share the single estimated value, and only 25.6% are fully measured.

6.1 Top, bottom and most frequent values

- **TCP retransmission.** The highest customer averages about 4.30 billion bytes and the lowest is 97 bytes. The most common value (55,974 customers) is the estimated average, not a repeated measurement.
- **RTT.** The highest is 48,462 ms and two customers average exactly 0 ms, both logging artefacts. The most common value, 62.9 ms (18,514 customers), is the estimated average.
- **Throughput.** The highest customer averages about 142,000 kbps. 108 customers average 0 kbps, meaning no data was recorded. The most common values (7.5 and 31.5 kbps) look like default connection rates.

The full top, bottom and most-frequent lists are in Appendix C.

So what: “Most frequent” lists for TCP and RTT show the estimated average, not customer behaviour. This is a limit of the brief’s mean-replacement rule, not a finding about the network.

6.2 Throughput and retransmission by handset

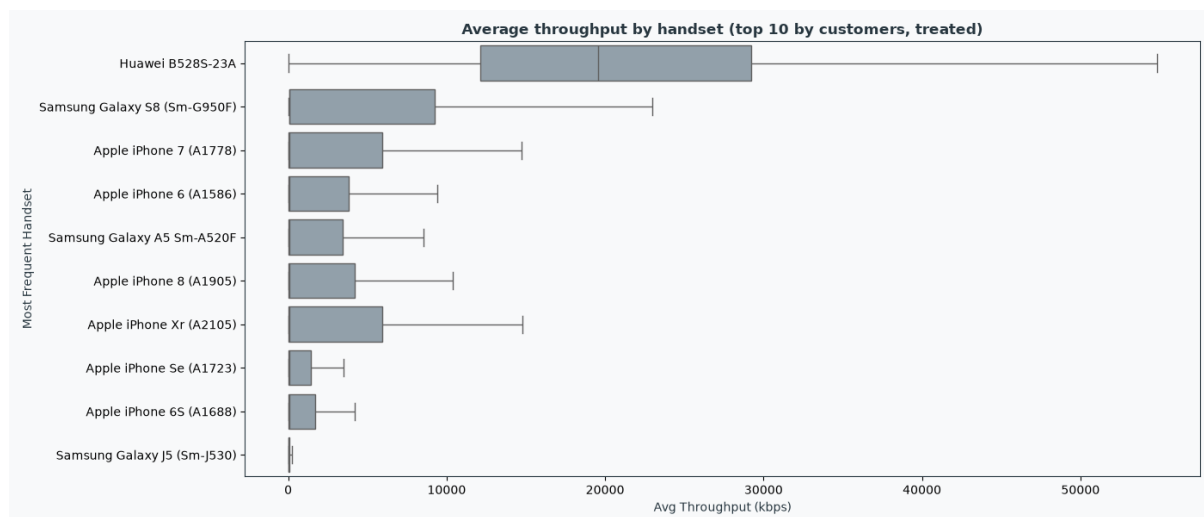


Figure 7. Average throughput by most frequent handset (ten most common, treated).

The Huawei B528S-23A router has by far the highest typical speed (median about 19,600 kbps). The Samsung Galaxy J5 has the lowest (median 39 kbps), and the other phones sit at medians of 54–68 kbps. Their upper quartiles run from about 1,400 to 9,200 kbps, so each phone group is mostly slow connections with a fast minority, and the median alone hides that. The router’s retransmission is also 5–14 times the phones’ among customers whose TCP was measured (median 4.97 million bytes against 0.35–0.93 million), though those medians rest on very different subsamples (84% of router customers against 7.5–20.4% of phone customers).

So what: Device type accounts for much of the throughput gap, mostly the router. Handset differences also mix in plan and location, which the data cannot separate.

6.3 Three experience segments (k-means, k = 3)

SEGMENT	CUSTOMERS	AVG TCP RETRANS. (MB)	AVG RTT (MS)	AVG THROUGHPUT (KBPS)	ROUTER USERS
GOOD EXPERIENCE	36,929 (34.6%)	4.5	38.4	9,090	19.5%
LOW THROUGHPUT	43,146 (40.4%)	20.9	22.4	628	2.1%
HIGH OR UNMEASURED LATENCY	26,781 (25.1%)	19.3	67.2	1,229	9.3%

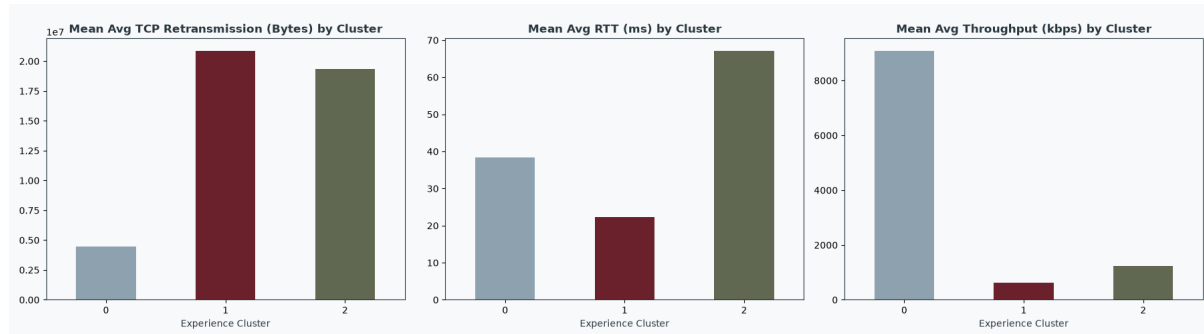


Figure 8. Average TCP retransmission, RTT and throughput by experience segment.

- **Good experience (34.6%).** Throughput is 7–14 times the other groups. Routers are over-represented: 68% of all router customers sit here, but 80.5% of the segment are not routers, so “good experience” is not simply “owns a router”.
- **Low throughput (40.4%, the largest).** Lowest throughput and lowest RTT. The data cannot say why: device or plan limits, cell load and application behaviour are all possible. Its TCP figure is mostly estimated and should be ignored.
- **High or unmeasured latency (25.1%).** Highest RTT, but 68.9% of its customers have RTT estimated in every session, and it holds about 99% of all such customers. It is largely a group whose latency was not measured.

How firm are these segments?

They are stable across random starts (agreement 0.999) but depend on preprocessing. Without TCP, agreement with the main solution drops to 0.53; with extremes capped instead of replaced, to 0.43. Only the contrast between good-throughput and low-throughput customers survives every version. On untreated data a very small group appears (167 customers, 0.16%) with average TCP retransmission of 2.19 billion bytes: either a real connection problem or a counter artefact. Mean replacement removes it by construction.

So what: Report throughput as the finding. There is no dominant poor-experience majority, but there are two different issues of similar size, a throughput gap and a latency question. Check the cause with cell-level, plan and device data before proposing any network spend.

7. Task 4: User satisfaction

Read this first

The file contains no measured satisfaction (no survey, complaint, churn or revenue data). The score required by the brief is a behavioural proxy: how far a customer sits from the least-engaged group and from the worst-experience group. We call it the “proxy score” where the difference matters.

7.1 How the scores are built

- **Engagement score:** distance from the least-engaged segment (Task 2). Higher means further from the low-engagement profile.
- **Experience score:** distance from the worst-experience segment (Task 3, the high or unmeasured latency group). Higher means lower RTT and retransmission and higher throughput.
- **Satisfaction (proxy) score:** the average of the two, each first rescaled to 0–1 so neither dominates. The overall average is 0.317 (range 0.02–0.93).

Both scores move as designed (engagement score against its three inputs: 0.57–0.84; experience score against throughput: +0.66, retransmission: –0.57, RTT: –0.41). These are consistency checks, not validation.

The experience score largely tracks whether RTT was measured: customers with RTT estimated in every session average 0.39 against 2.38 for fully measured customers.

7.2 Top 10 customers by proxy score

RANK	CUSTOMER (MSISDN)	ENGAGEMENT	EXPERIENCE	SATISFACTION (PROXY)
1	33659445167	5.721	4.708	0.932
2	33647092626	6.026	4.380	0.923
3	33769765653	6.188	4.183	0.916
4	33763740829	5.712	4.477	0.908
5	33699535294	5.478	4.526	0.894
6	33658271446	5.585	4.318	0.881
7	33661397206	4.704	4.987	0.879
8	33659324811	5.459	4.381	0.877
9	33623057467	5.176	4.524	0.869
10	33760919975	5.491	4.258	0.867

These are the highest-scoring customers under equal weighting. The overall ranking is robust to the weighting (rank agreement 0.92–0.98), but top-10 lists are not: only 7–9 of the ten names are shared between weightings. Any shortlist should state the weighting that produced it.

7.3 Two satisfaction segments (k-means, k = 2)

SEGMENT	CUSTOMERS	AVG ENGAGEMENT SCORE	AVG EXPERIENCE SCORE	AVG PROXY SCORE
ENGAGED, WEAKER EXPERIENCE	43,694 (40.9%)	2.33	1.21	0.295
LESS ENGAGED, BETTER EXPERIENCE	63,162 (59.1%)	1.07	2.55	0.331



Figure 9. Customers by engagement score and experience score, coloured by segment.

This is not a satisfied-versus-unsatisfied split. The two segments describe a trade-off between engagement and experience, and their average proxy scores are close because the two scores offset each other. Because the experience score is partly an estimation effect, the “weaker experience” group is partly a group with unmeasured latency.

7.4 Predicting the score

MODEL	TEST R ²	COMMENT
GRADIENT BOOSTING (SELECTED)	0.9921	MAE 0.0082, RMSE 0.0119; gap between train and test 0.0002; 5-fold cross-check 0.9923
RANDOM FOREST	0.9752	
LINEAR AND RIDGE REGRESSION	0.7949	Plateau: the score is a nonlinear (distance-based) function of its inputs
LASSO REGRESSION	0.7948	
MEAN-ONLY BASELINE	-0.0002	Floor

This is formula recovery, not prediction

The target is built from the same six inputs the model uses, so a flexible model can approximate it almost perfectly. The R^2 shows the score can be reproduced, not that it forecasts churn, revenue or real satisfaction. **Do not quote 0.992 as predictive skill.** A genuine predictive model needs an independent outcome such as churn, revenue or survey responses.

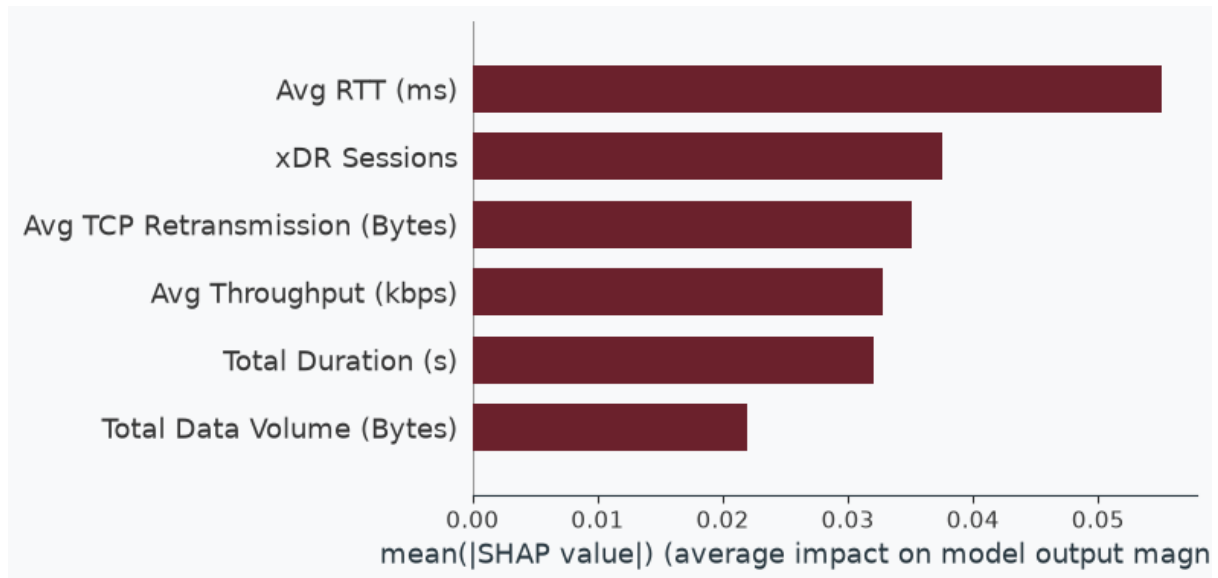


Figure 10. Importance of each input in reconstructing the proxy score (SHAP).

Three importance measures (SHAP, permutation and the model's own) agree on the leaders but not the order: RTT leads on two, sessions on the third, and no input dominates (largest-to-smallest ratio 2.5–4.8×). Total data volume ranks last on all three. RTT's lead partly reflects the block of customers sitting at the estimated RTT value, which the model uses as a flag, lowering their predicted score by about 0.10. **These importances explain how the model rebuilds the proxy, not what drives customer satisfaction.**

7.5 Traffic against proxy score: a priority segment

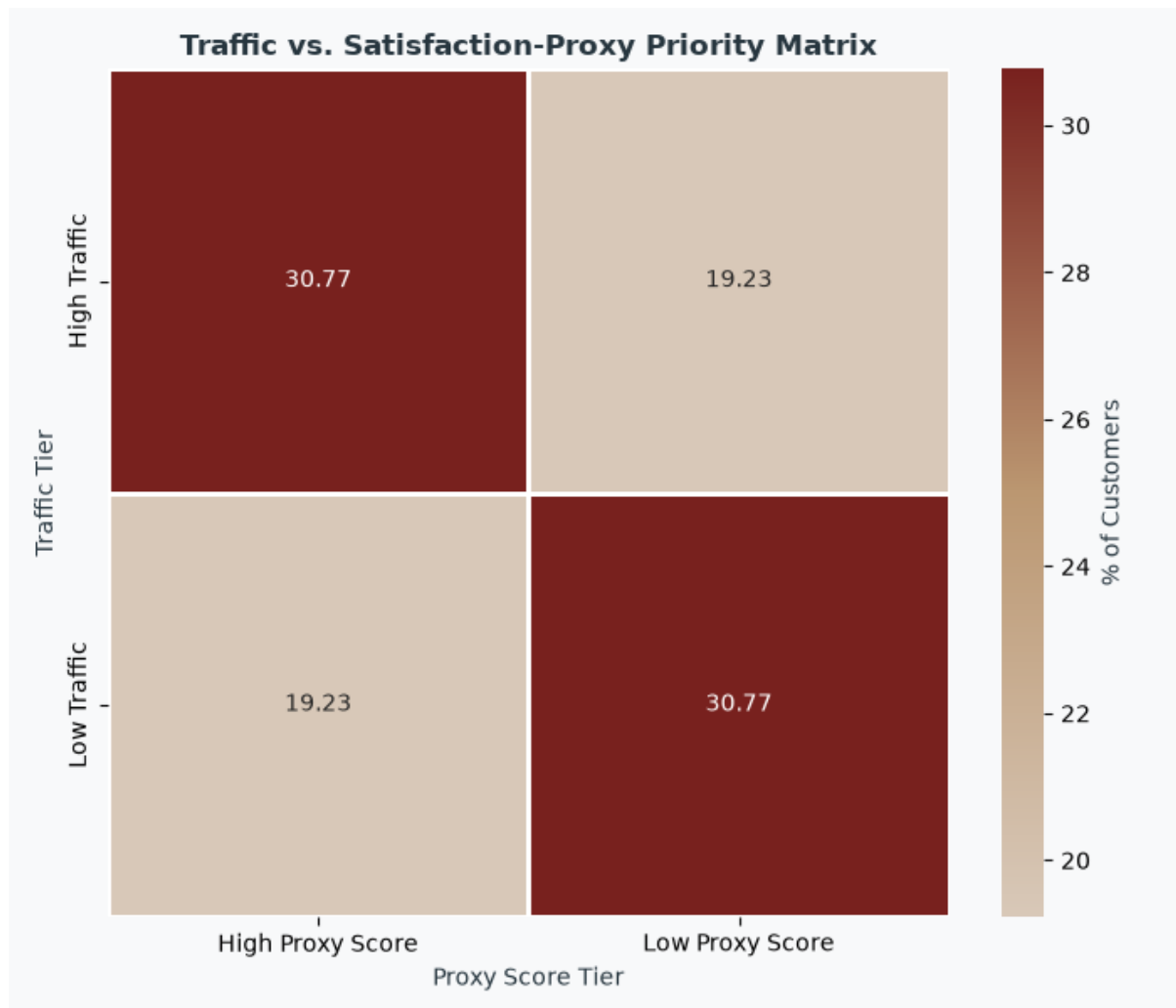


Figure 11. Share of customers by traffic tier and proxy-score tier (traffic split at the median).

20,548 customers (19.2%) have high traffic but a low proxy score. They carry 23.7% of treated traffic (25.5% untreated). This is a candidate segment to investigate, not a confirmed retention risk: 45.4% of its members are in the high or unmeasured latency group, and 34.2% have RTT estimated in every session. Under two alternative experience definitions, 81.6% and 71.4% of the segment stays in place, so most but not all of it survives.

So what: Test any retention offer for this segment against real outcome data. The file cannot show that heavy traffic means high value, or that a low proxy score means a customer is unhappy.

7.6 Export and model tracking

The scored customer table (customer ID, engagement, experience and proxy scores, segment labels, scoring version and timestamp) is validated and exported (one row per customer, no missing values, scores in range), with a CSV fallback. Each model run is logged with code version, start and end time, source, parameters, metrics (R^2 , MAE, RMSE), loss convergence (test loss within 1% of its final value after 198 of 200 trees) and the saved model file. A drift check flags any input whose average moves by more than a quarter of its reference standard deviation. A simulated +50% RTT shift confirms the alarm fires. This is file-

based tracking and a reference check, not a live monitoring service. The MySQL export and its SELECT evidence are included with the deliverables (Section 12).

8. The central insight: what the data can and cannot tell us

This is formula recovery, not prediction

The target is built from the same six inputs the model uses, so a flexible model can approximate it almost perfectly. The R^2 shows the score can be reproduced, not that it forecasts churn, revenue or real satisfaction. **Do not quote 0.992 as predictive skill.** A genuine predictive model needs an independent outcome such as churn, revenue or survey responses.

REASONABLY FIRM	HANDLE WITH CARE	CANNOT BE KNOWN FROM THIS FILE
- A base of 106,856 identifiable customers. - Apple, Samsung and Huawei hold 90% of sessions. - A home-router segment sits inside Huawei's count. - Throughput splits customers into good and low groups. - Traffic concentration is moderate, not extreme.	- Engagement segments: they change with the outlier treatment. - The latency group: largely unmeasured RTT. - TCP retransmission: mostly estimated. - Application shares: counters do not match totals. - The satisfaction proxy and its model.	- Revenue, margin, ARPU, or whether TellCo is undervalued. - Churn, growth or customer lifetime value. - Real customer satisfaction. - Why throughput is low. - Weekly cycles, billing cycles, trends or seasonality.

9. Growth potential and investment view

Assessment: cautiously positive on the behavioural evidence, unproven on the financials.

STRENGTHS	GROWTH LEVERS TO TEST	RISKS AND UNKNOWNNS
106,856 identifiable customers. - A High-engagement segment (22.8%) carrying about 37% of treated traffic. - Moderate concentration: not dependent on a thin slice of customers. - A distinct router segment (9.9% of customers) with strong throughput.	- Retain and upsell the High-engagement segment. - Investigate the 20,548 high-traffic, low-proxy-score customers. - Separate home-broadband plans for router customers. - Targeted trade-in tests on the iPhone 6S–8 base. - Improve throughput for the low-throughput group once the cause is known.	- No revenue, cost, ARPU or churn data. - One week of data: no trend or seasonality. 40.4% of customers have low throughput; the cause is unknown. - Missing TCP and RTT data may point to weak network monitoring at TellCo. - Counter definitions (Gaming, Other, totals) are unclear.

The investor's earlier success came from focusing on the most profitable segment. Here the equivalent candidates are the High-engagement segment and the router base. The data can identify them, but without revenue and churn data it cannot say whether either is the most profitable.

Recommendation: continue due diligence

The behavioural evidence **neither supports nor rules out a purchase**. It identifies questions for TellCo, not a valuation. Required before a buy / do-not-buy decision:

- 1. Financial data:** revenue, ARPU, cost, margin, churn and customer lifetime value. This is mandatory for any valuation.
- 2. A longer, evenly sampled window** (several weeks at least) and the extraction rule that produced this one.
- 3. Outcome data** (surveys, complaints, churn) to test the proxy and support a real predictive model.
- 4. Network data:** cell-level, location, plan and device data, plus an explanation for the missing RTT and TCP values, the 24-hour duration mass and the 32-bit-like TCP maximum.
- 5. Definitions** of the Gaming and Other counters, the total download and upload construction and the device gap.
- 6. Further analysis:** repeat segmentation and experience analyses on untreated or capped data, and split every experience result into routers and phones.

10. Prioritised recommendations

PRIORITY	RECOMMENDATION	WHY
HIGH	Request financial and churn data (revenue, ARPU, cost, margin, churn, lifetime value) before any valuation.	The usage data cannot show profitability, growth or retention.
HIGH	Obtain a longer, evenly sampled extract and the extraction rule.	One week of end-dated sessions cannot show trends, seasonality or churn.
HIGH	Rely on throughput, not TCP or RTT, for any investor-facing network metric.	Throughput is almost fully measured; TCP is largely estimated and RTT partly so.
MEDIUM	Analyse routers and phones separately in all network and marketing work.	Router customers inflate throughput and retransmission averages.
MEDIUM	Quote untreated figures for heavy-user and concentration questions.	Mean replacement hides 29.9% of raw traffic and understates concentration.
MEDIUM	Ask TellCo to define "Gaming" and "Other" and explain the total download and upload construction.	App shares and the Gaming story depend on definitions the file does not give.
MEDIUM	Pilot a retention test for the High-engagement and high-traffic, low-proxy-score groups, judged on real outcomes.	Both are candidates, not confirmed risks.
LOW	Test iPhone trade-in offers and a separate router proposition.	Hypotheses from the device mix, to be tested because there is no revenue data.
LOW	Test k = 2 against k = 3 engagement segments on business outcomes.	k = 2 fits the data better; k = 3 is the brief's choice.

11. Limitations

- **One week, not a month.** No trend, seasonality, weekly cycle or churn signal can be assessed.
- **No financial or outcome data.** No valuation, retention or “value” claim is possible.
- **Mean replacement.** The brief’s rule overwrites the heaviest customers, records customers with four or more sessions as one, understates concentration and weakens correlations. The segmentation and several relationships depend on it.
- **Estimated network data.** TCP (59–65% estimated per direction; 63.2% of customers fully estimated) and RTT (17.4% of customers fully estimated) shape the experience groups and the experience score.
- **Constructed satisfaction.** The score is a proxy, the regression recovers a formula, and the scalers and segment centres behind the score were fitted on all customers, including the test split. Rankings also depend on weighting and on the experience definition (18–29% of the priority segment changes).
- **Application counters.** They do not reconcile with total volume, and some have near-identical distributions.
- **Duration is connection-open time,** with a large 24-hour mass, so it partly reflects system behaviour.
- **Handset coverage is 94%.** The effect of the 6.0% unknown sessions on handset findings was not tested; the router segment is inferred from model names.
- **Segment causes cannot be identified** from three customer-level network aggregates.

12. Delivery status and next steps

DELIVERABLE	STATUS	NEXT STEP
DATA CLEANING AND FEATURE ENGINEERING	Done	Document the cleaning rules for TellCo’s data team.
FEATURE STORE (VERSIONED FILE)	Done	Reuse for later modelling.
CLUSTERING WITH VALIDATION; REGRESSION AND SHAP SCORED CUSTOMER TABLE; MODEL RUN LOG; DRIFT CHECK	Done	Validate on real outcome data when available.
MYSQL EXPORT AND SELECT SCREENSHOT	Done	Move to a hosted tracking tool if the model is deployed.
WEB DASHBOARD	Done	Re-run against TellCo’s production database when available.
SLIDE DECK	Done	Add live data once a longer data window is available.
REUSABLE PIP-INSTALLABLE PACKAGE	Done	Update after financial data is received.
COST-ADJUSTED MARGIN MODELLING	Done	Extend as new data arrives.
	Not possible yet	Requires revenue, cost and bill-of-materials data.
TIME-SERIES TRACKING	Not possible yet	Requires a longer window and a session date field covering more than one week.

References

- NextHikes IT Solutions (2026). User Analytics in the Telecommunication Industry: problem statement and field descriptions.
- TelCo xDR dataset, telecom_data (2).xlsx (150,001 session rows, sessions ending 24–30 April 2019).
- Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825–2830.
- McKinney, W. (2010). Data Structures for Statistical Computing in Python. Proceedings of the 9th Python in Science Conference.
- Lundberg, S. M. and Lee, S.-I. (2017). A Unified Approach to Interpreting Model Predictions. Advances in Neural Information Processing Systems, 30.
- Friedman, J. H. (2001). Greedy function approximation: a gradient boosting machine. Annals of Statistics, 29(5), 1189–1232.

Appendix A. Top 10 most engaged customers per application

Total data (download + upload) per application, in MB, untreated, over the one-week window.

Social Media | Google

#	Customer	MB	#	Customer	MB
1	33626320676	43.4	1	33626320676	152.2
2	33760536639	39.8	2	33625779332	142.3
3	33659725664	35.4	3	33614892860	128.0
4	33614892860	28.3	4	33760536639	123.2
5	33625779332	27.1	5	33659725664	116.5
6	33667163239	24.2	6	33786323068	110.3
7	33786323068	24.0	7	33675877202	109.9
8	33669068942	23.8	8	33667163239	105.0
9	33603127838	23.1	9	33761268199	97.1
10	33658490784	23.0	10	33698756430	91.9

Email | YouTube

#	Customer	MB	#	Customer	MB
1	33626320676	42.4	1	33625779332	453.0
2	33614892860	40.8	2	33760536639	396.3
3	33625779332	40.6	3	33614892860	394.4
4	33786323068	36.3	4	33626320676	374.5
5	33659725664	36.0	5	33675877202	317.4
6	33760536639	33.7	6	33667163239	315.2
7	33675877202	31.5	7	33627080969	308.8
8	33665460546	30.4	8	33760413819	303.2
9	33667163239	30.3	9	33698792269	302.7
10	33698792269	29.1	10	33603127838	284.1

Netflix | Gaming

#	Customer	MB	#	Customer	MB
1	33659725664	399.5	1	33614892860	7,749.4
2	33614892860	361.4	2	33760536639	7,461.0
3	33625779332	357.0	3	33625779332	7,326.7
4	33760536639	334.6	4	33675877202	6,970.6
5	33626320676	328.7	5	33626320676	6,887.6
6	33760413819	318.3	6	33659725664	6,725.6
7	33667163239	313.9	7	33666464084	6,646.3
8	33675877202	309.1	8	33760413819	6,268.6
9	33786323068	305.9	9	33664712899	6,103.9
10	33761268199	292.1	10	33698792269	5,753.7

Other

#	Customer	MB
1	33626320676	8,167.9
2	33614892860	7,639.3
3	33675877202	6,798.5
4	33625779332	6,354.6
5	33603127838	6,326.7
6	33659725664	6,317.4
7	33626948251	5,305.4
8	33627080969	5,117.8
9	33761268199	5,077.8
10	33658361927	5,013.7

Appendix B. Engagement segments: minimum, maximum, average and total

Non-normalised values on the treated table (the brief-mandated primary basis).

SEGMENT (CUSTOMERS)	METRIC	MIN	MAX	AVERAGE	TOTAL
LOW ENGAGEMENT (57,834)	Sessions	1	2	1	59,009
	Duration (seconds)	7,142	123,625	69,489	4,018,831,880
	Data (GB)	0.03	1.34	0.51	29,243
LONG-HOLD, LIGHT TRAFFIC (24,654)	Sessions	1	2	1	25,355
	Duration (seconds)	118,617	324,964	172,768	4,259,430,444
	Data (GB)	0.03	1.33	0.49	12,110
HIGH ENGAGEMENT (24,368)	Sessions	1	3	2	52,788
	Duration (seconds)	18,235	324,991	161,850	3,943,963,142
	Data (GB)	0.22	1.65	1.01	24,648

Segment totals are on the treated table, where the heaviest customers have been replaced by averages, so they understate real traffic.

Appendix C. Top, bottom and most frequent values (per customer)

Per-customer averages: top and bottom 10 and the most frequent values. TCP and RTT frequency tables are dominated by estimated averages.

TCP retransmission (bytes, average per customer)

#	TOP 10	BOTTOM 10	MOST FREQUENT VALUE	CUSTOMERS
1	4,301,477,439	97	21,650,430 (estimated)	55,974
2	4,288,120,860	128	20,885,480	367
3	4,268,647,350	129	20,885,510	167
4	4,254,659,643	134	767,578	145
5	4,211,257,819	143	21,267,950	82
6	4,166,594,973	176	20,885,500	81
7	4,137,937,737	176	768,908	78
8	4,117,804,973	177	766,340	78
9	3,968,120,596	179	766,286	70
10	3,786,870,933	182	20,886,840	65

Round-trip time (ms, average per customer)

#	TOP 10	BOTTOM 10	MOST FREQUENT VALUE	CUSTOMERS
1	48,462.0	0.0	62.93 (estimated)	18,514
2	27,424.0	0.0	14.5	2,950
3	16,167.5	1.0	19.5	2,367
4	13,639.0	2.0	19.0	1,639
5	13,150.0	2.5	20.0	1,583
6	12,857.5	3.0	15.0	1,577
7	12,694.0	4.0	14.0	1,456
8	12,369.0	4.5	24.5	1,308
9	11,505.3	4.5	20.5	1,207
10	10,490.0	5.0	15.5	1,190

Throughput (kbps, average per customer)

#	TOP 10	BOTTOM 10	MOST FREQUENT VALUE	CUSTOMERS
1	141,965.5	0.0	7.5	2,872
2	132,862.0	0.0	31.5	2,138
3	132,588.3	0.0	48.5	1,094
4	127,475.0	0.0	45.0	1,062
5	119,491.0	0.0	49.0	1,025
6	117,682.5	0.0	48.0	938
7	115,491.0	0.0	49.5	918
8	114,151.5	0.0	44.5	881
9	113,667.0	0.0	45.5	864
10	113,252.5	0.0	47.5	856

Appendix D. Plain-language notes on the statistics

TERM	MEANING IN THIS REPORT
PEARSON / SPEARMAN CORRELATION	How closely two measures move together (–1 to +1). Spearman compares rankings and is less sensitive to extreme customers.
GINI COEFFICIENT	How unevenly traffic is spread across customers: 0 means everyone uses the same, 1 means one customer uses everything.
K-MEANS CLUSTERING	A method that groups customers with similar values. We fix the number of groups (k) as the brief requires.
SEPARATION SCORE (SILHOUETTE)	How cleanly the groups separate; higher is better. 0.51 is moderate, 0.34 is weak.
AGREEMENT BETWEEN TWO GROUPINGS (ARI)	1 means identical groups, 0 means unrelated. 0.998 to 0.999 across random starts is very stable; 0.44 between treated and capped versions is low.
PCA (PRINCIPAL COMPONENT ANALYSIS)	Combines related measures into a few “factors”. The first factor here is overall usage.
R ² , MAE, RMSE	How closely model predictions match the target. R ² = 1 is a perfect match; MAE and RMSE are typical errors in score units.
SHAP	A way of showing how much each input pushes the model’s prediction up or down.
MEAN REPLACEMENT	The brief’s rule: values far above or below the typical range are replaced by the column average (computed without those values).
ESTIMATED (IMPUTED) VALUE	A missing measurement filled with the column average; it is not an observation.
TESTS USED	Kruskal–Wallis (volume differs across duration deciles: moderate effect, 0.07 treated, 0.16 untreated) and Mann–Whitney pairwise tests with multiple-testing correction (42 of 45 pairs differ). With over 100,000 customers, tests are nearly always significant, so effect sizes matter more than p-values.